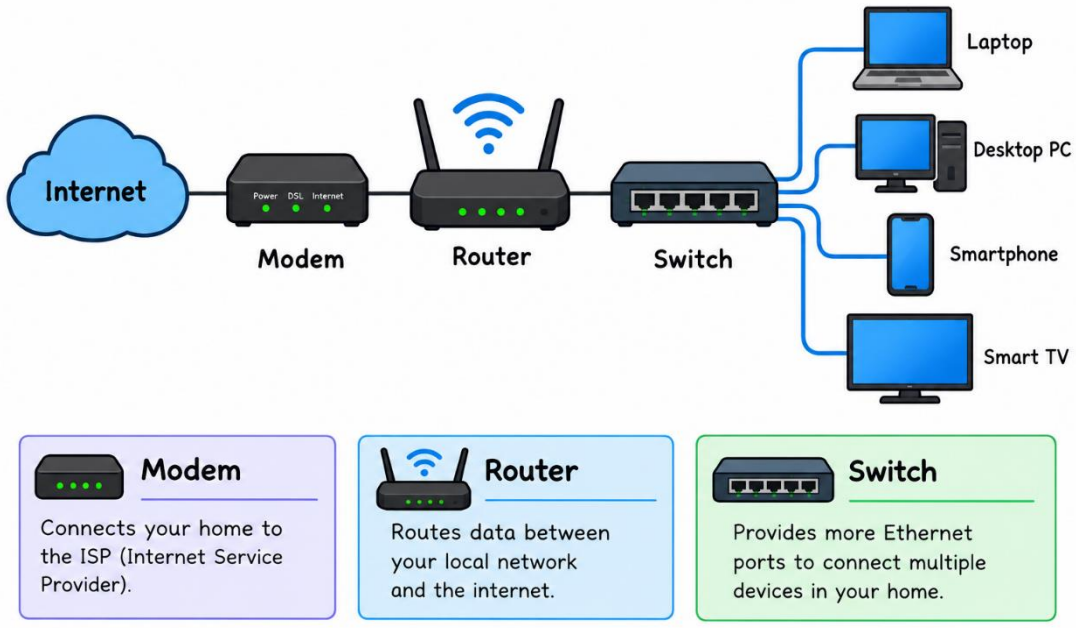
Routers/modems/switches

Tasks

1. Draw a simple network diagram showing how a router, modem, and switch connect together in a typical home setup. Label each device and explain the role of each in one line on your diagram.

Ans.:-



**2. Log in to your home WiFi router's admin panel (usually at 192.168.0.1 or 192.168.1.1) and find the section that shows connected devices. List the names or MAC addresses of at least 3 devices currently connected to your network.

Hint: If you don't have access to a physical router, search online for a router admin demo page and use that for this task.**

Ans.:-

Steps :-

1. Connect your PC/laptop to your home Wi-Fi.
2. Open Chrome/Edge.
3. Enter one of these addresses:
    - 192.168.1.1
    - 192.168.0.1
4. Log in with your router's admin username/password.
5. Find Connected Devices, Attached Devices, Client List, DHCP Clients, or Device List.
6. Note at least 3 device names or MAC addresses.

No.	*Connected Device*	*MAC Address*
1	Laptop	XX:XX:XX:XX:XX:XX
2	Smartphone	XX:XX:XX:XX:XX:XX
3	Smart TV	XX:XX:XX:XX:XX:XX

3. Create a table comparing routers, modems, and switches. Include at least 4 points for each: main function, where it is used (home/office), whether it assigns IP addresses, and if it supports WiFi.

Ans.:-

Point	**Router**	**Modem**	**Switch**
Main Function	Connects different networks and routes data between them	Connects your local network to the ISP/Internet	Connects multiple devices within the same LAN
Where Used	Home and office	Home and office	Mainly offices, schools, and data centers
Assigns IP Addresses?	Yes, usually through DHCP	No	No
Supports Wi-Fi?	Yes, if it is a wireless router	Usually No (unless it is a modem-router combo)	No, standard switches use Ethernet cables
Example	Connects PCs/phones to the Internet	Connects your network to the ISP	Connects PCs, printers, and servers together

**4. Imagine you are setting up a small gaming café. Choose and specify which devices (router, modem, switch) you would need, how many of each, and briefly explain the reason for each choice.

Hint: Consider the number of computers, internet connection, and need for wired vs wireless access.**

Ans.:-

Small Gaming Cafe Setup :-

For a small gaming café with 10 gaming computers, I would use the following devices:

Device	Quantity	Reason
Modem	1	Connects the café's network to the ISP/Internet connection.
Router	1	Manages the network, assigns IP addresses using DHCP, and provides Internet access to all computers.
Switch	1 × 16-port	Connects the 10 gaming PCs using Ethernet cables. Extra ports are available for printers, servers, or future PCs.

Network Layout :-

Internet/ISP - Modem - Router - 16-Port Switch - 10 Gaming PCs

Wi-Fi: If customers also need Wi-Fi for phones, I would use a wireless router or add 1 wireless access point. For gaming PCs, wired Ethernet is preferred because it generally provides more stable and lower-latency connectivity.